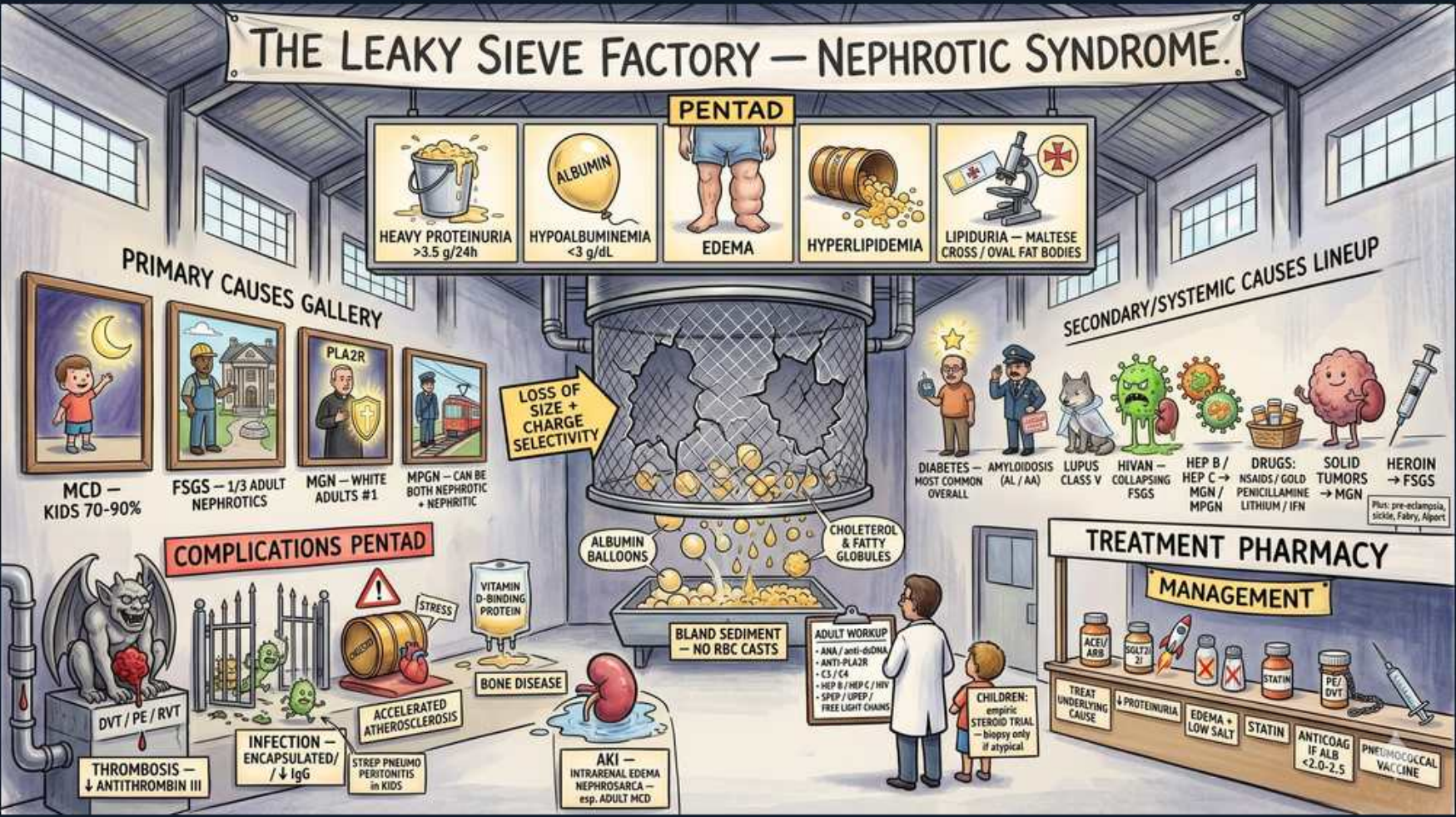


Glomerular Disease — Nephrotic Syndrome (Leaky Sieve Factory)



1. Heavy proteinuria >3.5 g/day

WHAT YOU SEE

Pentad — bucket overflowing >3.5 g/24h

WHAT IT ENCODES

The defining lesion of nephrotic syndrome is loss of the glomerular filtration barrier's charge and size selectivity, dumping >3.5 g protein/24h into the urine. The barrier has three layers: fenestrated endothelium, GBM (heparan sulfate gives the negative charge that repels albumin), and podocyte slit diaphragm. On urine dipstick this shows as 3+/4+ albumin; a spot urine protein:creatinine ratio >3-3.5 g/g approximates 24h collection. Watch for frothy/foamy urine on history — a classic exam buzzword for heavy proteinuria.

BOARD TRAP

WRONG answer: calling 1-2 g/day 'nephrotic.' Sub-nephrotic proteinuria (<3.5 g/day) with hematuria and RBC casts is the NEPHRITIC range — discriminator is the magnitude of protein loss plus an active vs bland sediment.

2. Hypoalbuminemia

WHAT YOU SEE

Pentad — deflating balloon <3 g/dL

WHAT IT ENCODES

Urinary albumin loss outpaces hepatic synthesis, dropping serum albumin <3 g/dL (often <2.5). Low albumin reduces plasma oncotic pressure (each 1 g/dL albumin contributes ~5.5 mmHg), which is the engine for edema and the trigger for compensatory hepatic lipoprotein synthesis. It also means fewer protein-binding sites, so the free (active) fraction of highly protein-bound drugs and total serum calcium readings shift — correct calcium upward ~0.8 mg/dL per 1 g/dL albumin drop.

BOARD TRAP

WRONG answer: expecting elevated or normal albumin. Hyperalbuminemia is never nephrotic; if a 'nephrotic' vignette shows normal albumin with hypertension and RBC casts, reclassify it as nephritic — the low-albumin/edema/lipid triad is what locks in nephrotic.

3. Edema

WHAT YOU SEE

Pentad — swollen legs

WHAT IT ENCODES

Pitting edema is driven primarily by reduced oncotic pressure pulling fluid into the interstitium, compounded by primary renal Na⁺ retention (overflow/underfill mechanisms). Distribution is gravity- and tissue-laxity dependent: periorbital edema on waking (kids), then dependent leg/sacral edema and anasarca; can progress to ascites and pleural effusions. Manage with dietary Na⁺ restriction (<2 g/day) and loop diuretics (e.g., furosemide), often needing higher doses because the drug is albumin-bound and tubular albumin sequesters it.

BOARD TRAP

WRONG answer: attributing nephrotic edema mainly to hypertension/volume overload. That mechanism describes NEPHRITIC edema; nephrotic edema is oncotic-driven (hypoalbuminemia) — the discriminator is normal/low BP with profound hypoalbuminemia vs HTN with active sediment.

4. Hyperlipidemia

WHAT YOU SEE

Pentad — oil barrels

WHAT IT ENCODES

Low oncotic pressure stimulates hepatic synthesis of lipoproteins (increased LDL, VLDL, total cholesterol, often triglycerides) and urinary loss of regulatory factors like LCAT reduces lipid clearance. The result is atherogenic dyslipidemia that raises long-term cardiovascular risk. Treat with a statin (e.g., atorvastatin 10-40 mg daily) alongside the proteinuria-lowering and dietary measures.

BOARD TRAP

WRONG answer: hypolipidemia/low cholesterol. Lipids RISE in nephrotic syndrome because the liver overproduces lipoproteins — the discriminator from a malabsorption/cachexia vignette is the co-existing heavy proteinuria and edema, not weight loss.

5. Lipiduria — oval fat bodies

WHAT YOU SEE

Pentad — microscope, Maltese cross

WHAT IT ENCODES

Filtered lipoproteins are reabsorbed by tubular cells that slough into urine as oval fat bodies; free lipid droplets and fatty casts also appear. Under polarized light, cholesterol esters create a 'Maltese cross' birefringence pattern — a pathognomonic nephrotic clue. The sediment is otherwise bland (no dysmorphic RBCs, no RBC casts).

BOARD TRAP

WRONG answer: RBC casts as the urinary hallmark. RBC casts and dysmorphic RBCs are NEPHRITIC (glomerular bleeding); the nephrotic discriminator is fatty casts/oval fat bodies with a Maltese-cross under polarized light and an otherwise bland sediment.

6. MCD — kids 70-90%

WHAT YOU SEE

Primary causes gallery — child, moon/lipoid

WHAT IT ENCODES

Minimal change disease is the #1 cause of nephrotic syndrome in CHILDREN (70-90%). Light microscopy is normal, immunofluorescence is negative, and EM shows diffuse podocyte foot-process effacement; it is often triggered by recent viral infection/atopy or, rarely, Hodgkin lymphoma/NSAIDs. Albuminuria is highly selective. Treatment is empiric prednisone (e.g., 60 mg/m²/day) — most children are steroid-responsive, so biopsy is usually deferred unless atypical or steroid-resistant.

BOARD TRAP

WRONG answer: naming MCD the top ADULT primary cause. In adults the leading primary cause is FSGS (or membranous in white adults); MCD is only ~10-15% of adult nephrotic — the discriminator is patient age plus steroid responsiveness.

7. FSGS — adults 1/3

WHAT YOU SEE

A framed portrait in the Primary Causes Gallery labeled FSGS marked '1/3 ADULT NEPHROTIC' — the gallery frame casts it as the leading primary nephrotic cause in adults (focal, segmental scarring).

WHAT IT ENCODES

Focal segmental glomerulosclerosis causes ~1/3 of adult nephrotic syndrome and is the leading primary cause in adults overall, especially in patients of African ancestry (APOL1 risk alleles), with HIV (collapsing variant), obesity, or heroin use. Light microscopy shows focal (some glomeruli) and segmental (part of the tuft) sclerosis; EM shows foot-process effacement. It is often steroid-resistant and a major driver of progression to ESKD.

BOARD TRAP

WRONG answer: calling FSGS the top pediatric cause. In kids that is MCD; the FSGS discriminators are African ancestry, HIV/APOL1, obesity, partial steroid response, and segmental sclerosis on biopsy.

8. MGN — white adults #1

WHAT YOU SEE

A framed portrait labeled MGN tagged 'WHITE ADULTS #1' in the causes gallery — the frame flags membranous nephropathy as the top primary nephrotic cause in older/white adults (anti-PLA2R, subepithelial spikes).

WHAT IT ENCODES

Membranous nephropathy is the most common PRIMARY nephrotic cause in white/older adults. Primary disease is driven by anti-PLA2R antibodies (~70%); secondary causes include hepatitis B, SLE (class V), solid tumors (lung/colon), and drugs (NSAIDs, gold, penicillamine). Light microscopy shows thickened capillary loops; IF shows granular subepithelial IgG/C3; EM shows subepithelial deposits with 'spike-and-dome' GBM reaction. It carries the highest risk of renal vein thrombosis.

BOARD TRAP

WRONG answer: confusing membranous with membranoproliferative GN. MPGN shows a 'tram-track' double-contour GBM and often presents with a mixed nephritic-nephrotic picture and low complement; membranous shows subepithelial spikes, normal complement, and a pure nephrotic picture.

9. Secondary / systemic causes

WHAT YOU SEE

Secondary causes lineup — diabetes, lupus, amyloid

WHAT IT ENCODES

Systemic disease causes nephrotic syndrome more often than any single primary lesion: diabetic nephropathy is the SINGLE MOST COMMON cause overall (Kimmelstiel-Wilson nodules), followed by amyloidosis (AL/AA, apple-green Congo red birefringence), SLE (class V membranous), hepatitis B/C, NSAIDs, and solid tumors; heroin and HIV produce FSGS. Always screen the systemic context (glucose/A1c, SPEP/free light chains, ANA, viral serologies) before assuming a primary lesion.

BOARD TRAP

WRONG answer: jumping to a primary glomerulopathy in a long-standing diabetic with retinopathy and gradually rising proteinuria. Diabetes is the most common cause overall — the discriminator is the diabetic timeline (microalbuminuria years before overt proteinuria) plus retinopathy, which usually obviates biopsy.

10. Hypercoagulable / thrombosis

WHAT YOU SEE

Complications — DVT/PE, renal vein thrombosis gargoyle

WHAT IT ENCODES

Urinary loss of antithrombin III (plus increased hepatic fibrinogen and platelet activation) creates a hypercoagulable state. The classic complication is renal vein thrombosis — present with flank pain, hematuria, and worsening proteinuria — most strongly associated with membranous nephropathy; DVT and PE also occur. Prophylactic or therapeutic anticoagulation is considered when albumin is very low (e.g., <2-2.5 g/dL) or in high-risk membranous disease.

BOARD TRAP

WRONG answer: predicting a bleeding diathesis. Nephrotic patients CLOT, they do not bleed — the discriminator is antithrombin III loss; sudden flank pain + hematuria + LDH rise in a membranous patient should trigger 'renal vein thrombosis,' not stone or glomerular bleeding alone.

11. Infection risk

WHAT YOU SEE

Complications — encapsulated organisms

WHAT IT ENCODES

Urinary loss of immunoglobulins (especially IgG) and complement factor B, plus immunosuppressive therapy, predisposes to infection by ENCAPSULATED organisms. The board-classic is spontaneous bacterial peritonitis from *Streptococcus pneumoniae* in a child with ascites; also *E. coli* and *Haemophilus influenzae*. Pneumococcal vaccination is indicated.

BOARD TRAP

WRONG answer: assuming nephrotic patients are immunologically protected or that infection implies a separate primary immunodeficiency. The discriminator is acquired Ig loss in the urine — a nephrotic child with fever and abdominal pain plus ascites = pneumococcal peritonitis until proven otherwise.

12. Bland sediment — no RBC casts

WHAT YOU SEE

Lab tube — bland sediment, no RBC casts

WHAT IT ENCODES

Pure nephrotic syndrome has a BLAND urinary sediment: heavy proteinuria, fatty casts/oval fat bodies, and Maltese crosses, but no dysmorphic RBCs and no RBC casts. This contrasts sharply with nephritic syndrome's active sediment. A mixed sediment (proteinuria PLUS RBC casts) suggests a nephritic-nephrotic overlap such as lupus nephritis or MPGN.

BOARD TRAP

WRONG answer: reporting an active sediment with RBC casts as nephrotic. RBC casts indicate a NEPHRITIC/proliferative process (glomerular inflammation) — the discriminator is bland vs active sediment, and a mixed picture should prompt thinking of lupus or MPGN.

13. Treatment — ACEi/ARB + cause-specific

WHAT YOU SEE

Treatment pharmacy — statins, ACEi, anticoag, steroids

WHAT IT ENCODES

Universal nephrotic management lowers intraglomerular pressure and addresses complications: ACE inhibitor or ARB (e.g., lisinopril, losartan) to reduce proteinuria, a statin for dyslipidemia, dietary Na⁺ restriction with loop diuretics for edema, and anticoagulation when thrombotic risk is high. Cause-specific therapy is then layered on: empiric steroids for MCD, immunosuppression (steroids ± calcineurin inhibitors, rituximab for anti-PLA2R membranous) for primary lesions, and tight glucose/RAAS control for diabetic nephropathy.

BOARD TRAP

WRONG answer: giving high-dose steroids first-line for every nephrotic patient. Steroids are first-line for steroid-sensitive MCD, but membranous, FSGS, diabetic, and amyloid nephropathy require different (or no) immunosuppression — the discriminator is the underlying etiology, so empiric steroids without diagnosis is wrong in adults.
